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Machine learning classification of structural systems using exclusively geospatial and remote sensing attributes: A case study in Santo Domingo, Dominican Republic

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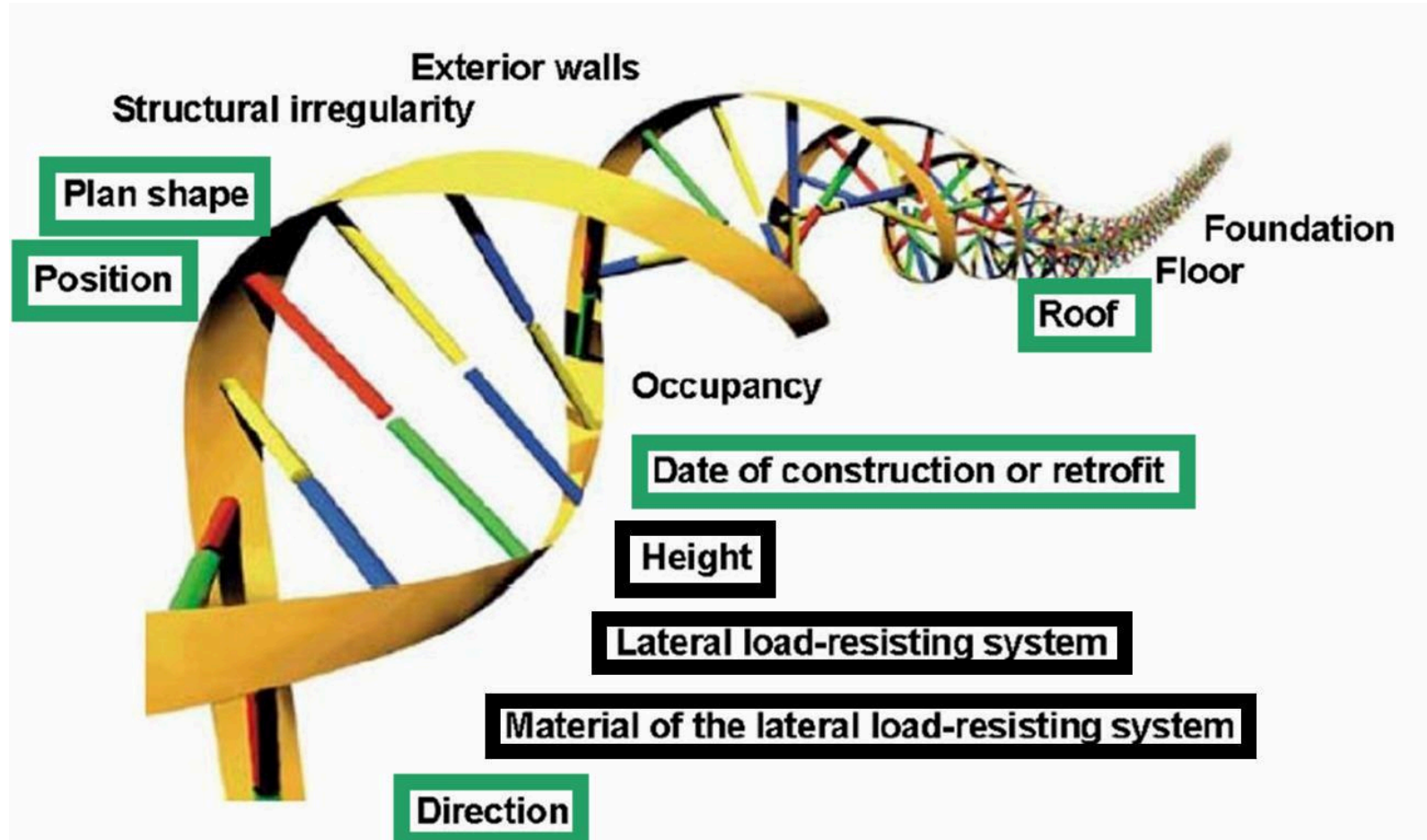
⁶ Departamento de Ingeniería y Gestión Forestal, ETSIMFYMN, Universidad Politécnica de Madrid, Spain

⁷ Centro de I+D+i en Infraestructuras Inteligentes y Sostenibles (CIVILis), ETSICCP, Universidad Politécnica de Madrid, Spain

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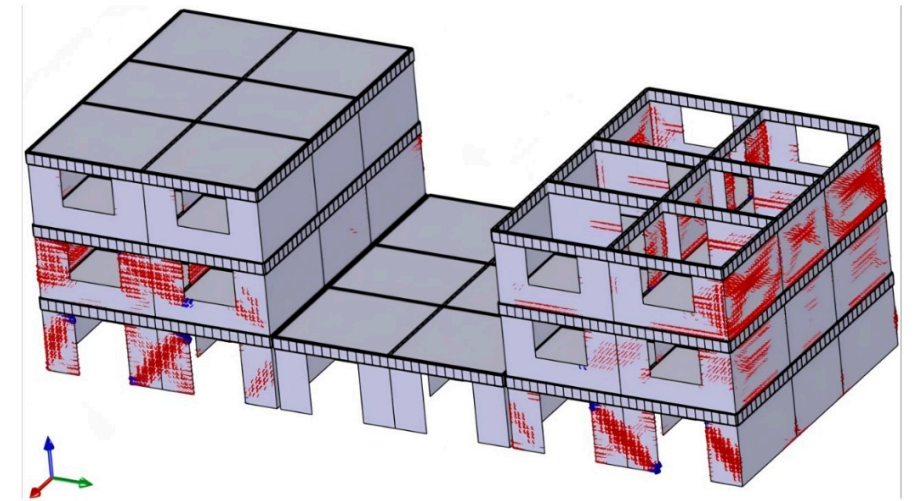
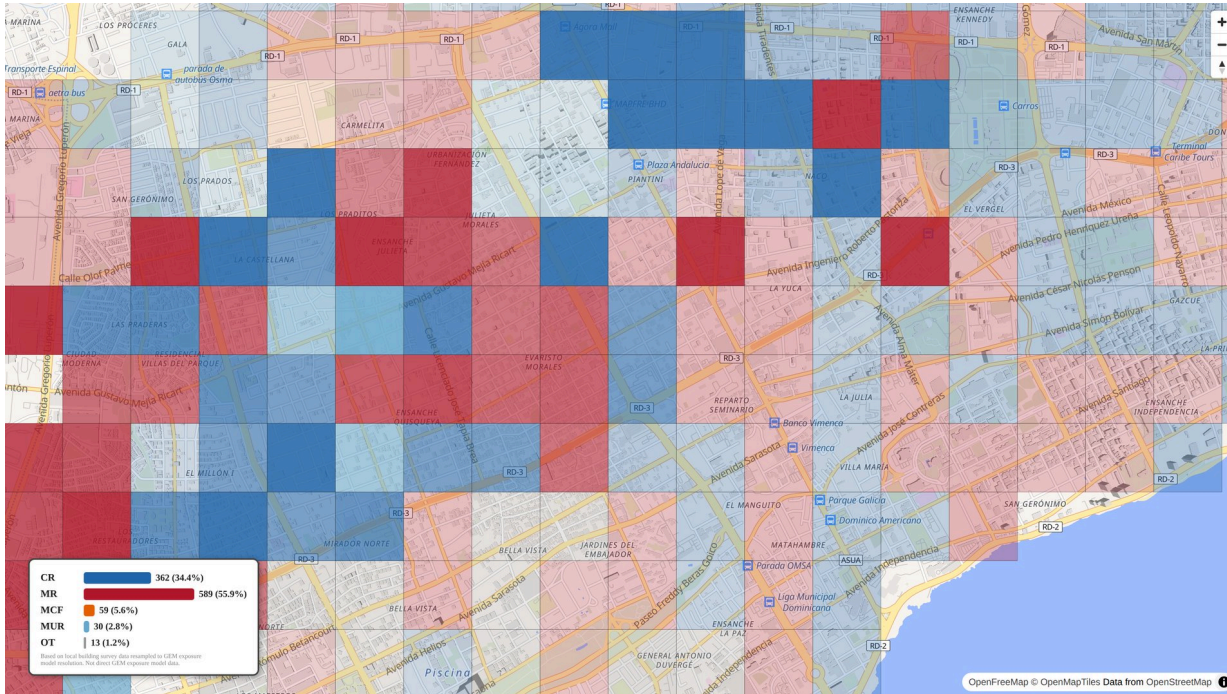
[linkedin.com/in/miguel-urena-pliego](https://www.linkedin.com/in/miguel-urena-pliego) github.com/MiguelUrenaPliego

Background: GEM attributes



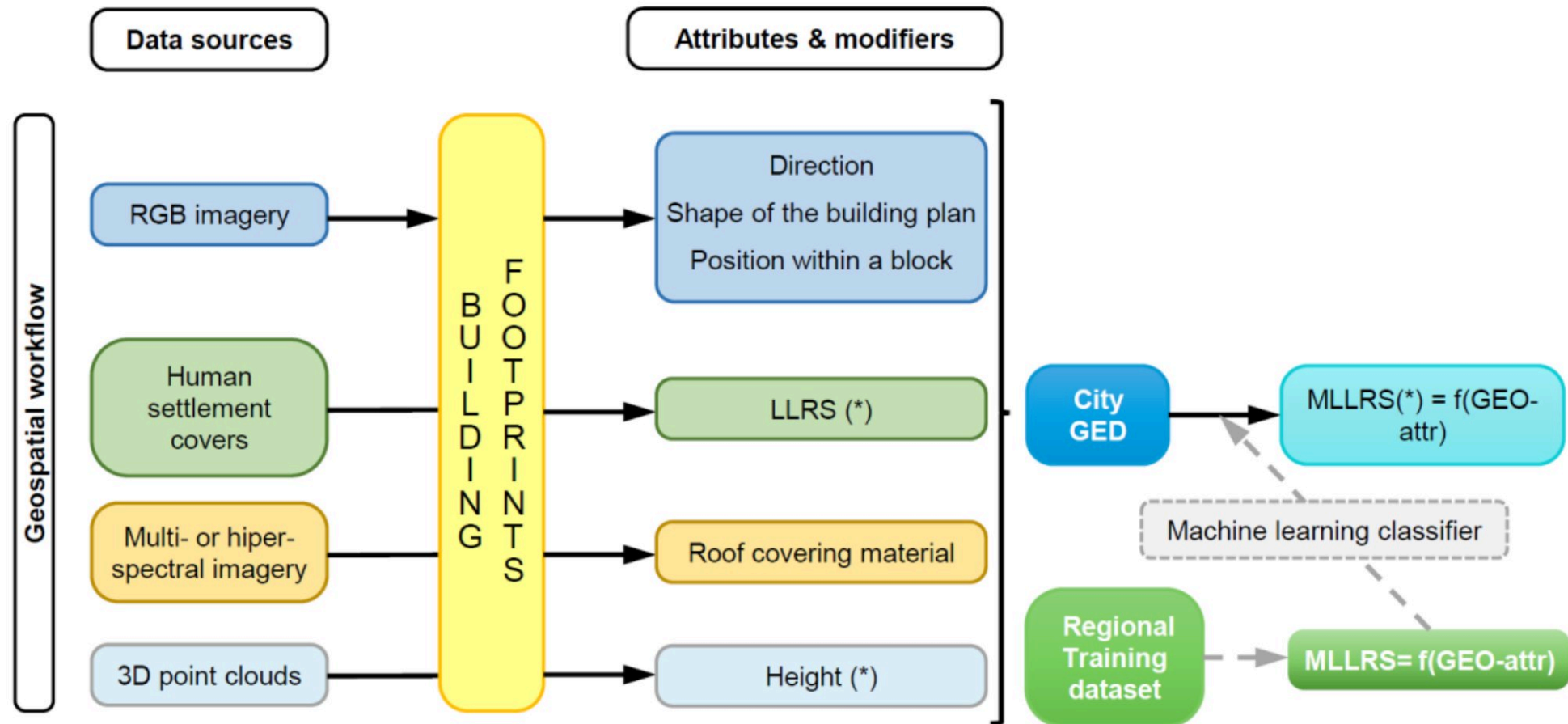
Brzev et al., 2013

Background: GEM exposure models

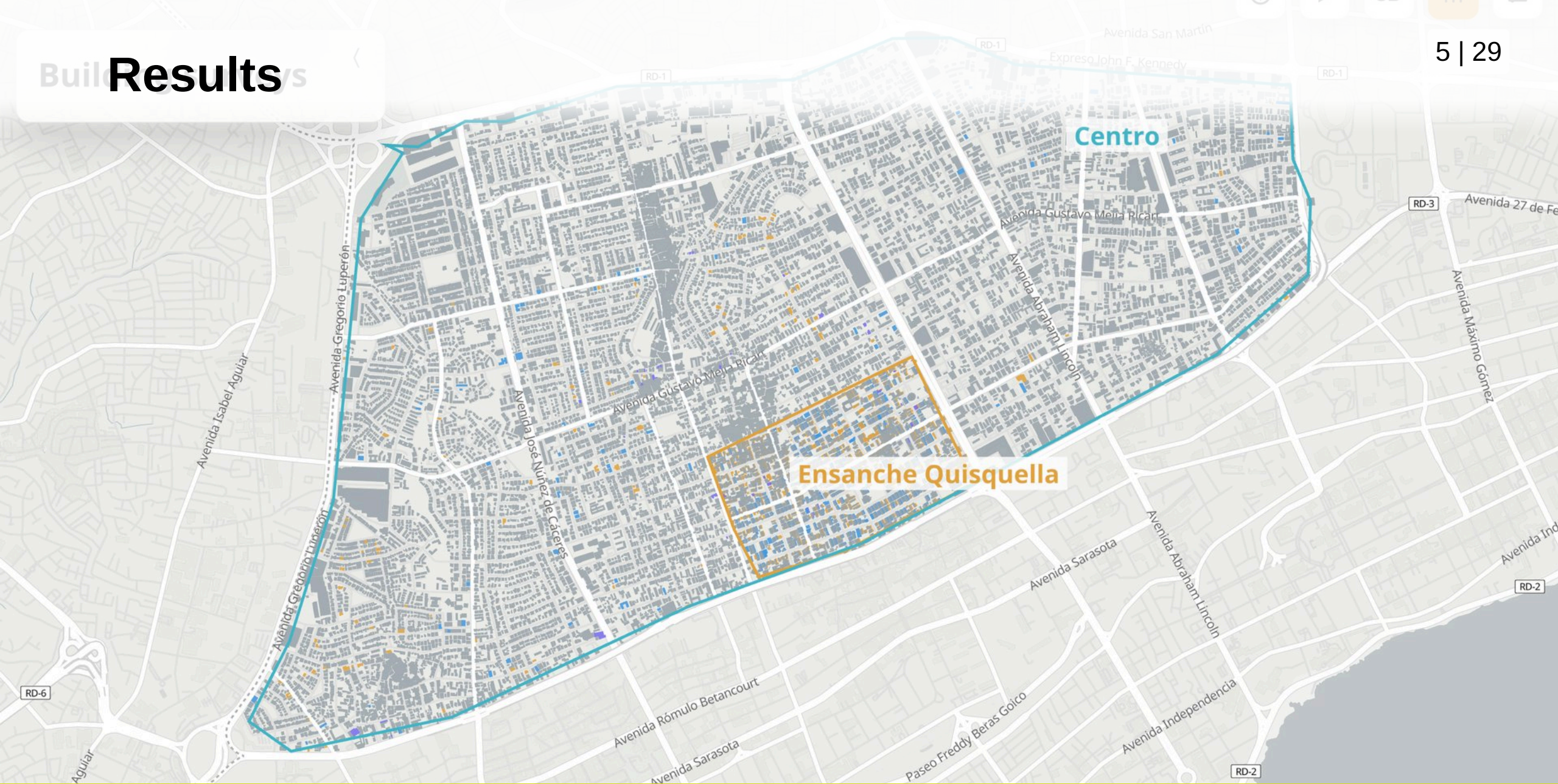


GEM Foundation, n.d.

Methodology: Proposed workflow



Yepes-Estrada et al., 2023



Ensanche Quisquella — Roof material

Roof material

- Metallic
- Bright concrete
- Asphalt
- Dark / shadowed
- Unlabeled

Surveyed coverage

Surveyed	Quisquella	821 (56%)
	Centro	232 (2%)
Not surveyed	Quisquella	643 (44%)
	Centro	13673 (98%)

Roof material

Metallic	Quisquella	474 (32%)
	Centro	12969 (93%)
Bright concrete	Quisquella	291 (20%)
	Centro	215 (2%)
Asphalt	Quisquella	423 (29%)
	Centro	96 (1%)
Dark / shadowed	Quisquella	276 (19%)
	Centro	625 (4%)
Unlabeled	Quisquella	0 (0%)
	Centro	0 (0%)

Footprint geometry

Metric	Mask2Former	SAM2	Microsoft
AJ	0.580	0.630	0.099
SBD	0.560	0.737	0.124
PQ	0.446	0.530	0.002
mAP	0.224	0.277	0.0003
sAP	0.370	0.526	0.044



Santo Domingo GT



Mask2Former



SAM2



Microsoft

Ureña-Pliego et al., 2026

Height

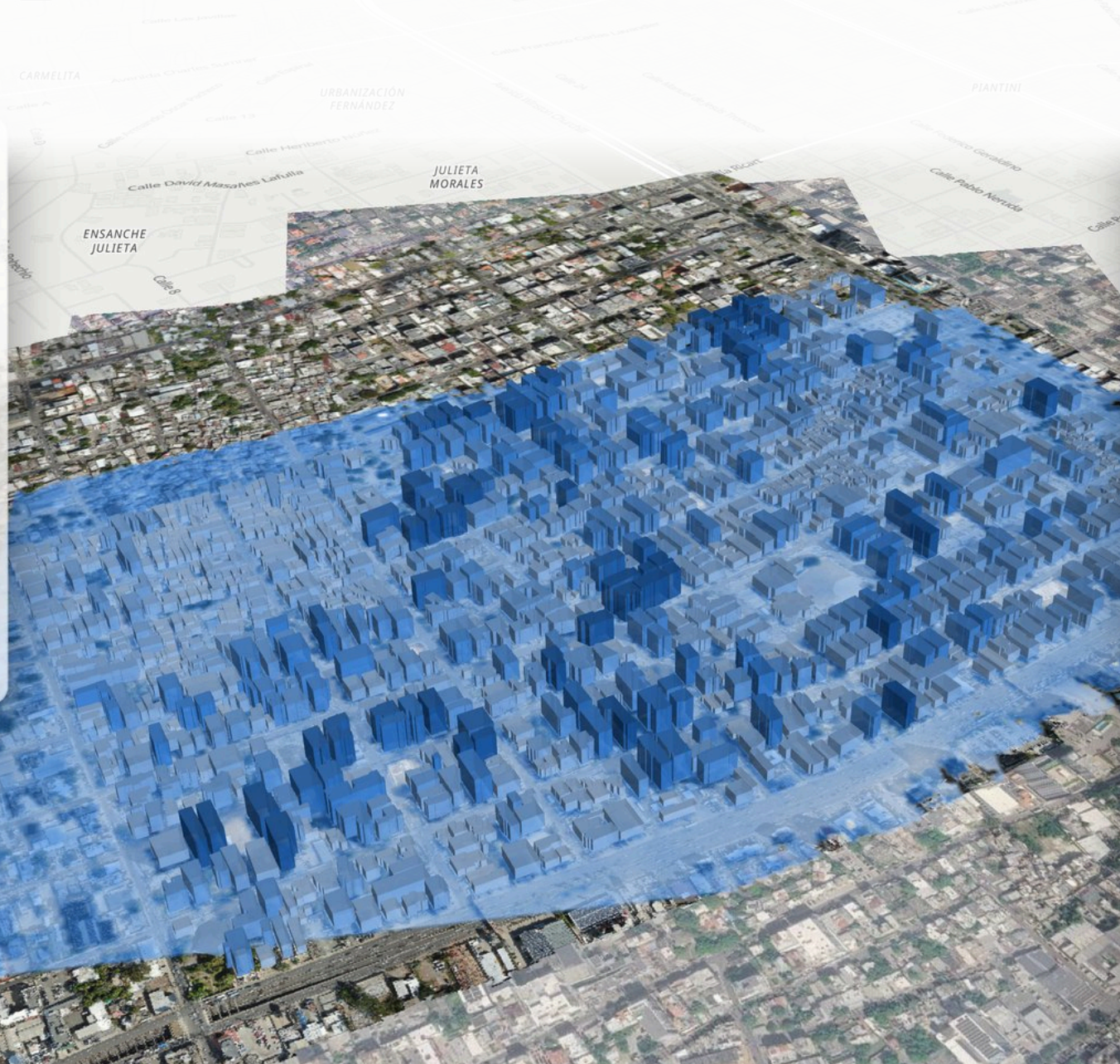
Ensanche Quisquilla

Height (m)

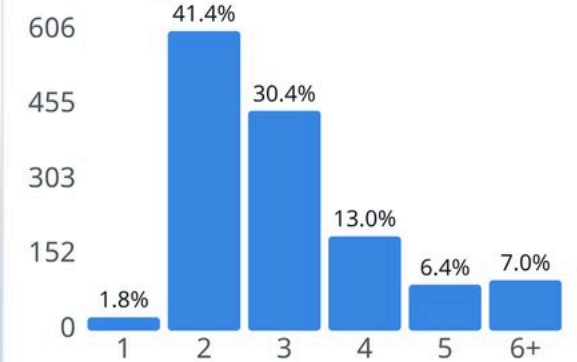
- 0.0 m
- 5.0 m
- 10.0 m
- 15.0 m
- 20.0 m

Height DSM-DTM

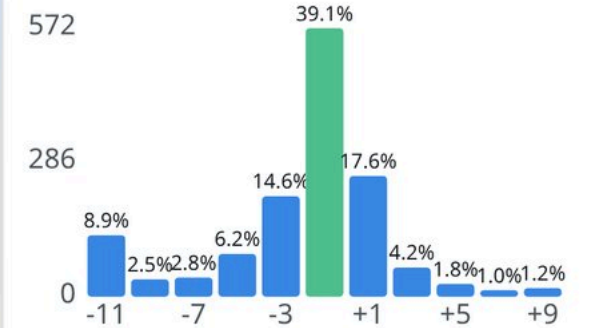
0 m 20 m



Buildings by floor count



Height error (vs survey)



RMSE: 5.74 m

Zhu et al., 2025

Height error (vs. survey)

Ensanche
Quisquilla

Height error (vs survey)
(m)

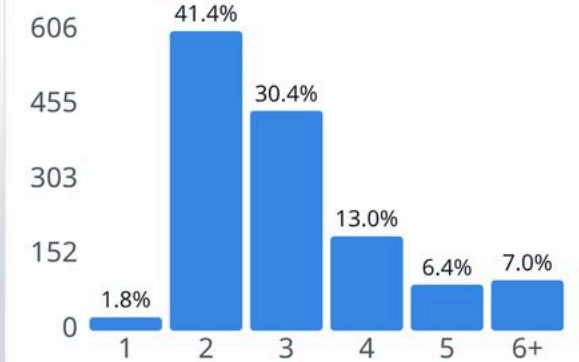
- -10.0 m
- -5.0 m
- +0.0 m
- +5.0 m
- +10.0 m

Height DSM-DTM

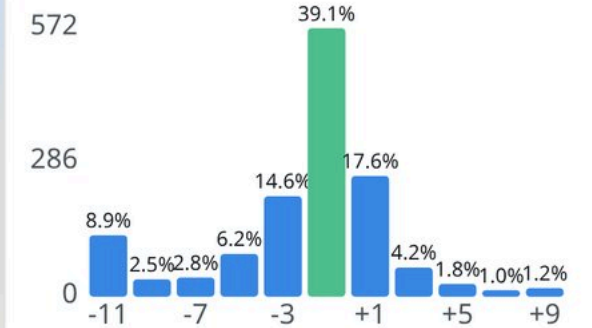
0 m 20 m



Buildings by floor count



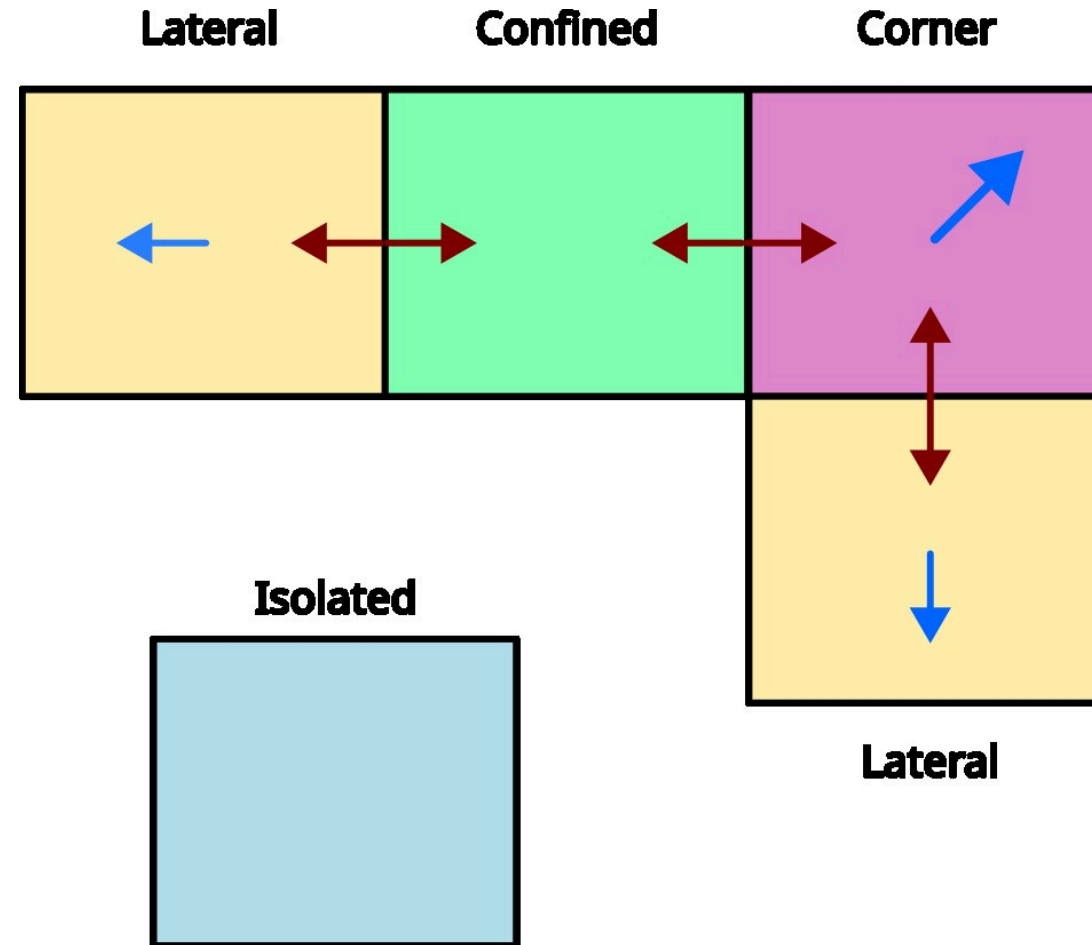
Height error (vs survey)



RMSE: 5.74 m

Zhu et al., 2025

Relative position within a block



Ureña-Pliego et al., 2026

Relative position within a block

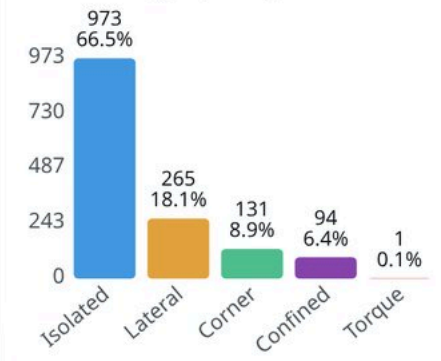
Ensanche Quisquella

Relative position

- Isolated (973)
- Lateral (265)
- Corner (131)
- Confined (94)
- Torque (1)

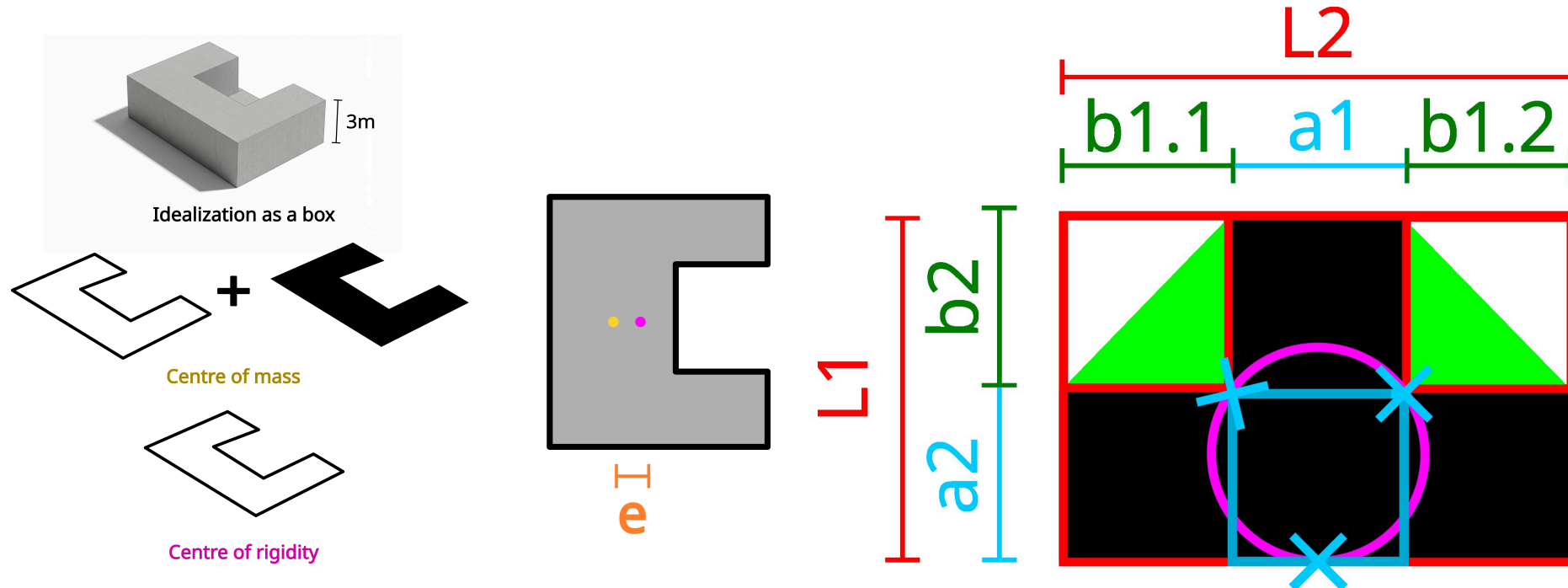


Buildings per position



Ureña-Pliego et al., 2026

Footprint shape



Ureña-Pliego et al., 2026

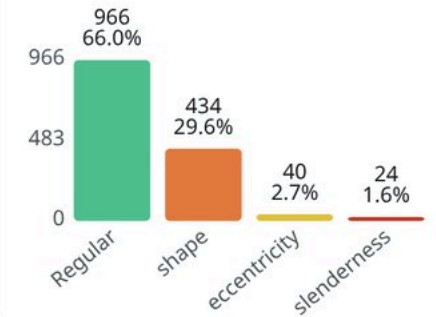
Footprint shape

Ensanche Quisquella

Shape index

- Regular
- Irregular: shape
- Irregular: eccentricity
- Irregular: slenderness

Shape index



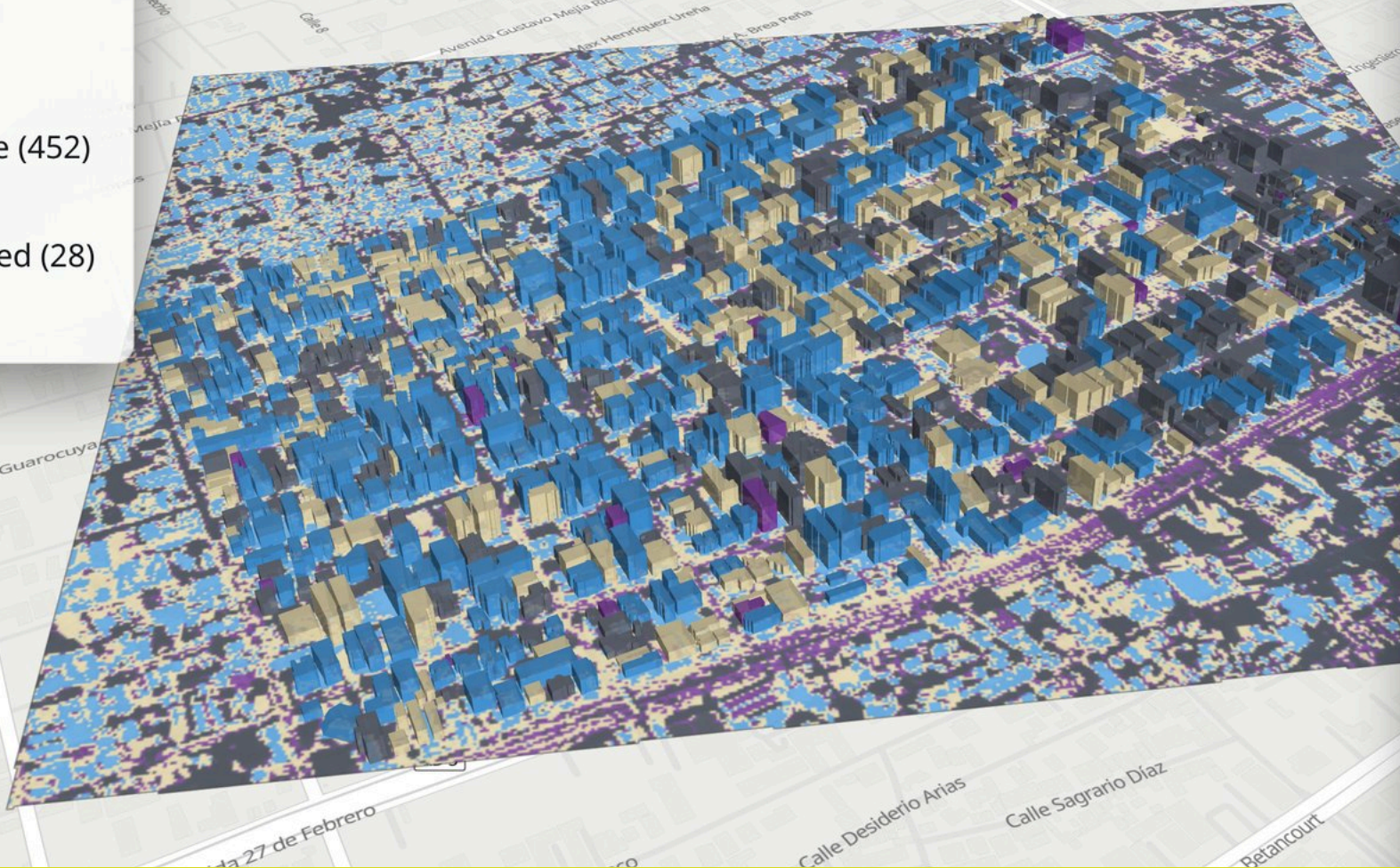
Ureña-Pliego et al., 2026

Roof cover material

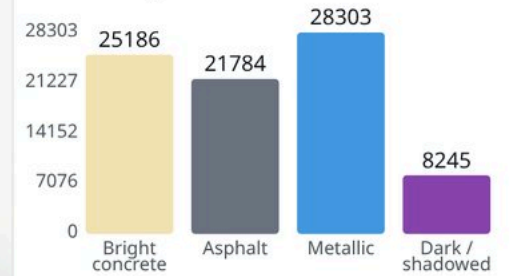
Ensanche Quisquilla

Roof material

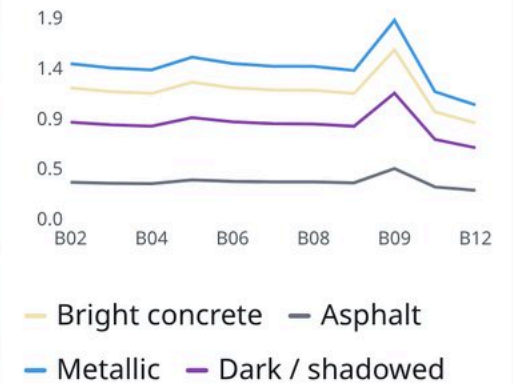
- Metallic (548)
- Bright concrete (452)
- Asphalt (428)
- Dark / shadowed (28)
- Unlabeled (8)



Pixels per cluster



Band cluster centroids



Torres et al., 2023

Construction or modification year: First construction year

Ensanche
Quisquilla

First construction year

- 1975
- 1979
- 1982
- 1986
- 1989

Buildings per year



Marconcini et al., 2021

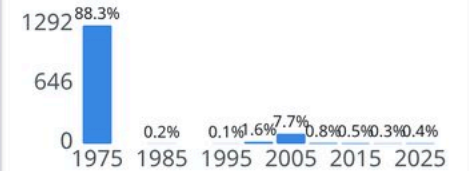
Construction or modification year: Last modification year

Ensanche
Quisquilla

Last modification year

- 1999
- 2006
- 2013
- 2019
- 2026

Buildings per year



Marconcini et al., 2021

Construction or modification year: Code quality

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Quisquealla

Code quality

- Pre-code (1292)
- Low code (141)
- Medium code (31)

Code quality mapping (Dominican Republic)

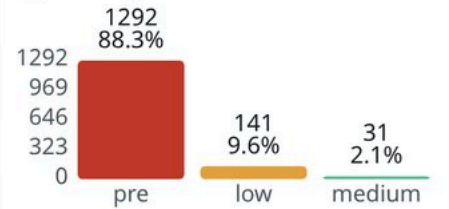
Effective year Code quality

Before 1979 Pre-code

1979 – 2010 Low code

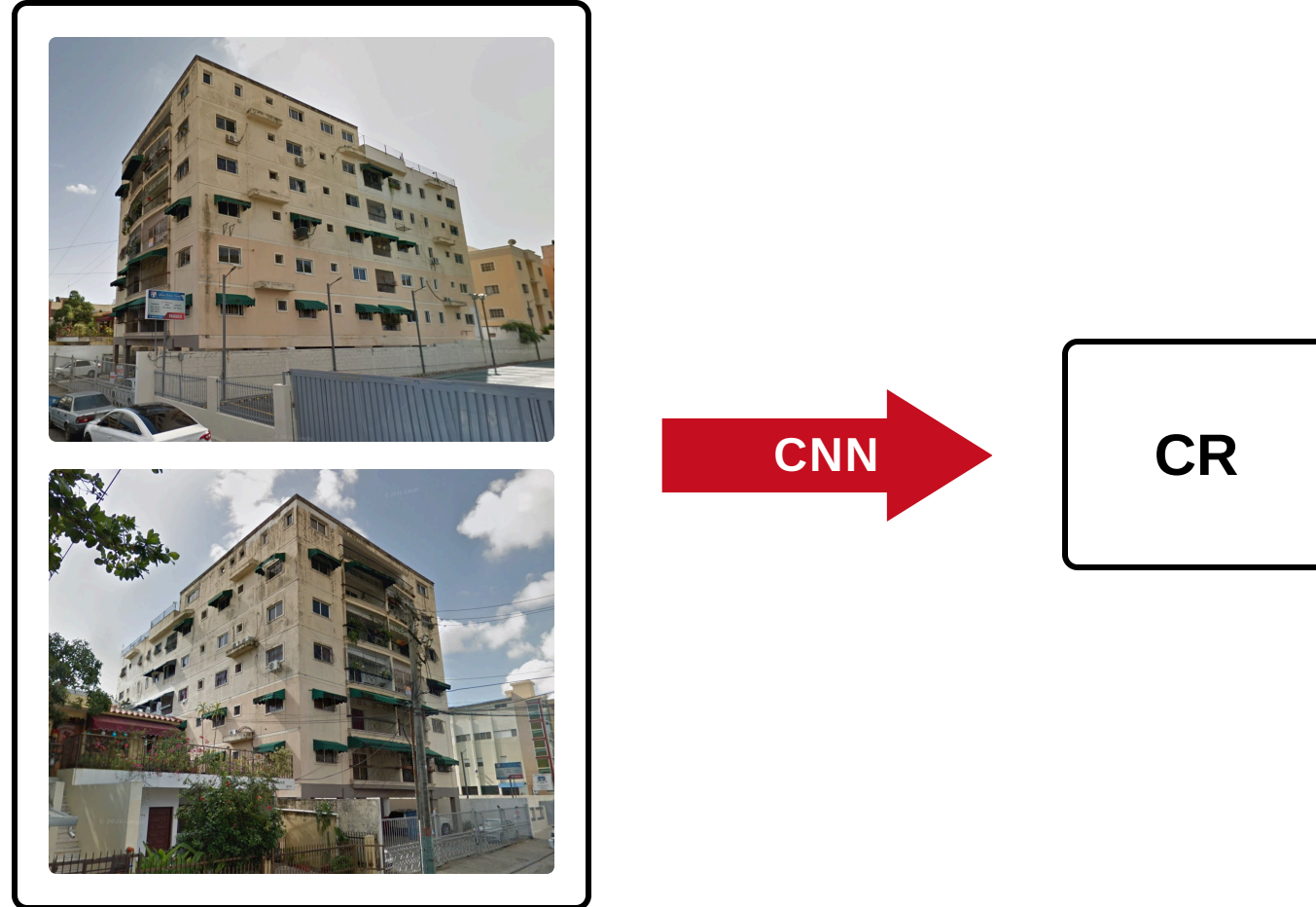
2011 onward Medium code

Buildings per code quality



Marconcini et al., 2021

Structural system



Lopes et al., 2023

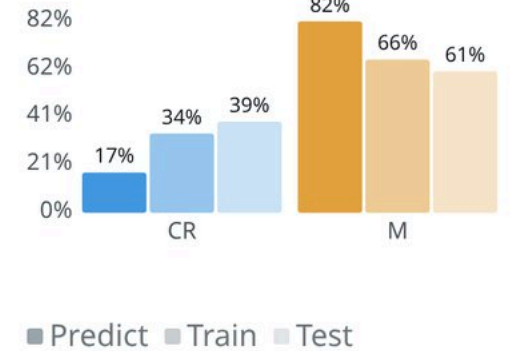
Structural system: Datasets

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Train / val / test split

- Train (572)
- Validation (108)
- Test (137)
- Inference (no split)

Structural systems



Structural system: Metrics (ground truth)

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Structural system (ground truth)

- Concrete (CR) (284)
- Masonry (MR | MCF) (535)
- Steel frames (S) (2)
- Unlabeled (643)



Confusion matrix

		Predicted	
		CR	M
CR	CR	230 81%	54 19%
	M	7 1%	526 99%

Evaluation metrics

Model	F1 $\pm 95\%CI$	Acc	K
Ensemble	0.91	0.93	0.83
CatBoost	0.89 $\pm 5\%$	0.91	0.79
XGBoost	0.89 $\pm 5\%$	0.90	0.78
HistGB	0.89 $\pm 5\%$	0.90	0.78
RandForest	0.88 $\pm 5\%$	0.89	0.76
LogReg	0.87 $\pm 5\%$	0.88	0.73

Structural system: Metrics (predicted)

Ensanche Quisquella

Structural system (predicted)

- Concrete (CR) (254)
- Masonry (MR | MCF) (1205)
- Unlabeled (5)



Confusion matrix

		Predicted	
		CR	M
Actual	CR	230 81%	54 19%
	M	7 1%	526 99%

Evaluation metrics

Model	F1 $\pm 95\%CI$	Acc	K
Ensemble	0.91	0.93	0.83
CatBoost	0.89 $\pm 5\%$	0.91	0.79
XGBoost	0.89 $\pm 5\%$	0.90	0.78
HistGB	0.89 $\pm 5\%$	0.90	0.78
RandForest	0.88 $\pm 5\%$	0.89	0.76
LogReg	0.87 $\pm 5\%$	0.88	0.73

Structural system: Metrics (error)

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Error (vs ground truth)

- Correct
- Incorrect
- No ground truth

Confusion matrix

		Predicted	
		CR	M
CR	CR	230 81%	54 19%
	M	7 1%	526 99%

Evaluation metrics

Model	F1 $\pm 95\%CI$	Acc	K
Ensemble	0.91	0.93	0.83
CatBoost	0.89 $\pm 5\%$	0.91	0.79
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HistGB	0.89 $\pm 5\%$	0.90	0.78
RandForest	0.88 $\pm 5\%$	0.89	0.76
LogReg	0.87 $\pm 5\%$	0.88	0.73

Structural system: Metrics (uncertainty)

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Quisquella

Uncertainty

Confident (0.0)

Uncertain (1.0)

Confusion matrix

		Predicted	
		CR	M
CR	CR	230 81%	54 19%
	M	7 1%	526 99%

Evaluation metrics

Model	F1 $\pm 95\%CI$	Acc	K
Ensemble	0.91	0.93	0.83
CatBoost	0.89 $\pm 5\%$	0.91	0.79
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HistGB	0.89 $\pm 5\%$	0.90	0.78
RandForest	0.88 $\pm 5\%$	0.89	0.76
LogReg	0.87 $\pm 5\%$	0.88	0.73

Structural system: Metrics (consensus)

 **Ensanche Quisquella**

Consensus (ensemble only)

 Low agreement All models agree

Confusion matrix

		Predicted	
		CR	M
CR	CR	230 81%	54 19%
	M	7 1%	526 99%

Evaluation metrics

Model	F1 $\pm 95\%CI$	Acc	K
Ensemble	0.91	0.93	0.83
CatBoost	0.89 $\pm 5\%$	0.91	0.79
XGBoost	0.89 $\pm 5\%$	0.90	0.78
HistGB	0.89 $\pm 5\%$	0.90	0.78
RandForest	0.88 $\pm 5\%$	0.89	0.76
LogReg	0.87 $\pm 5\%$	0.88	0.73

Structural system: Explainability

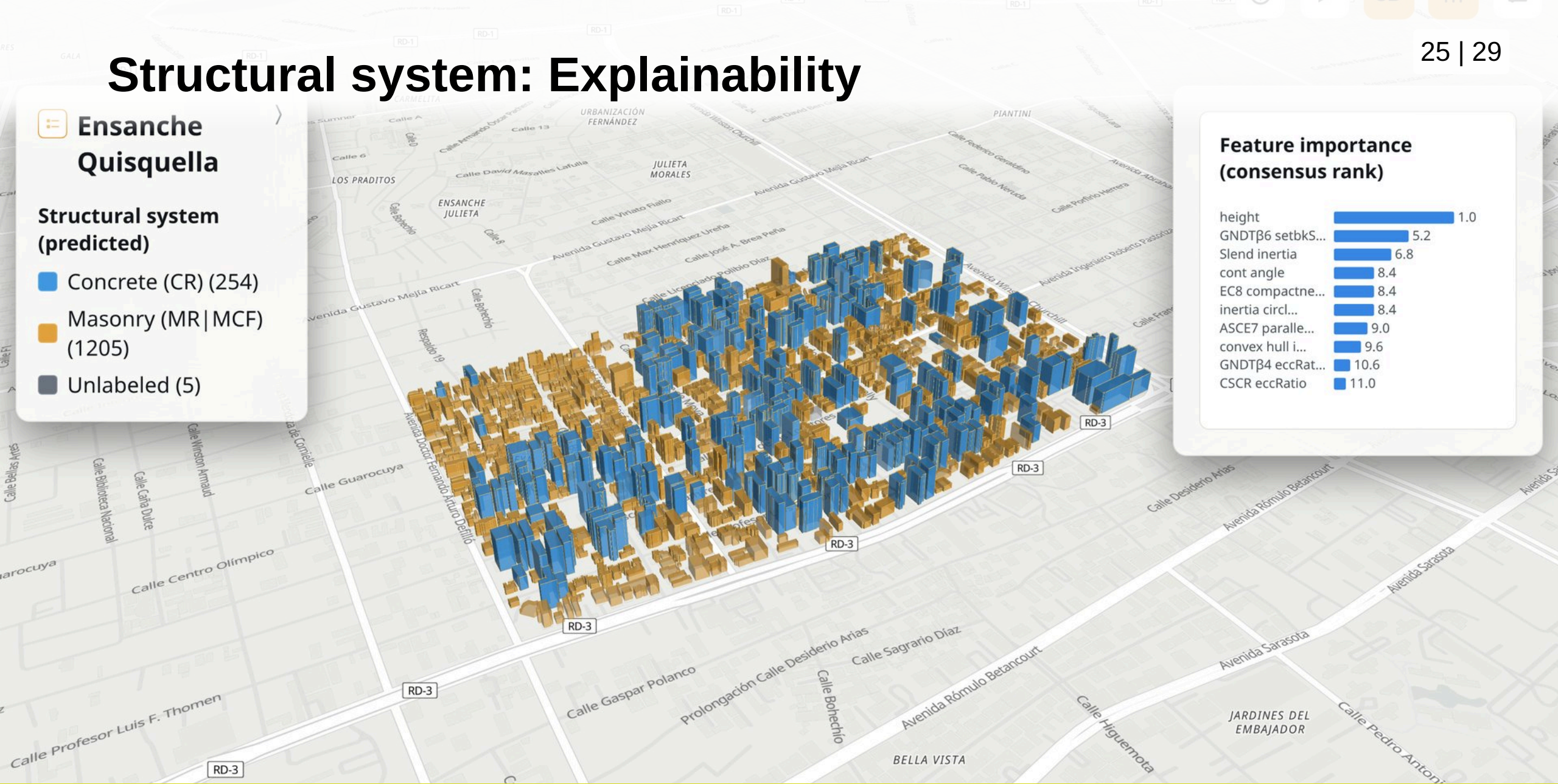
Ensanche Quisquella

Structural system (predicted)

- Concrete (CR) (254)
- Masonry (MR | MCF) (1205)
- Unlabeled (5)

Feature importance (consensus rank)

height	1.0
GNDTβ6 setbkS...	5.2
Slend inertia	6.8
cont angle	8.4
EC8 compactne...	8.4
inertia circl...	8.4
ASCE7 paralle...	9.0
convex hull i...	9.6
GNDTβ4 eccRat...	10.6
CSCR eccRatio	11.0



Structural system: Generalizability

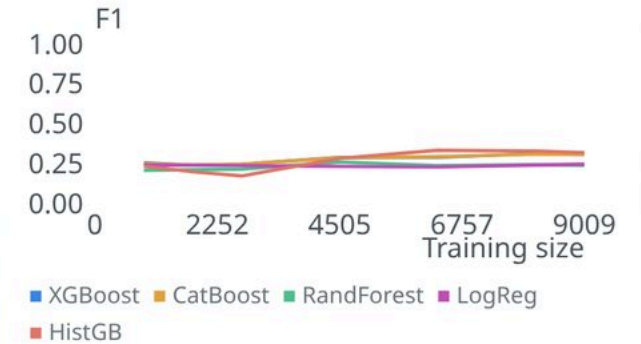
Ensanche Quisquella

Structural system (predicted)

- Concrete (CR) (449)
- Masonry (MR | MCF) (985)
- Adobe (ADO) (22)
- Steel frames (S) (3)
- Unlabeled (5)



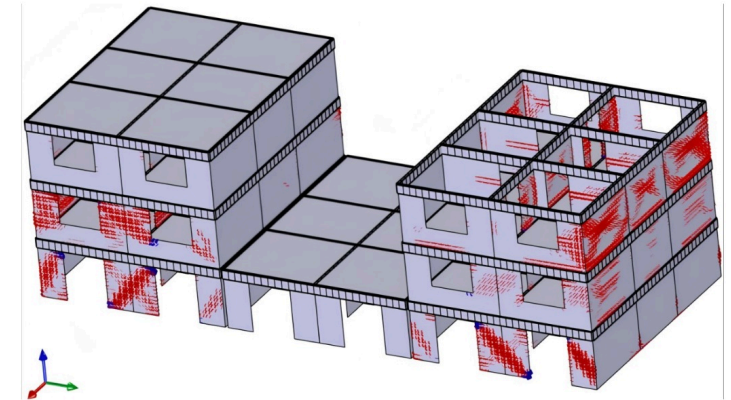
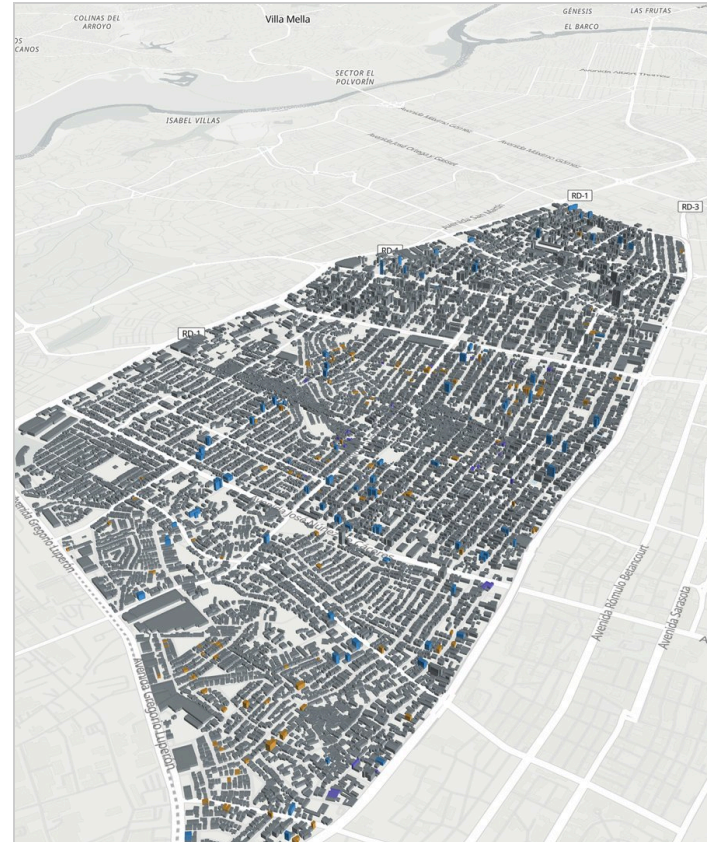
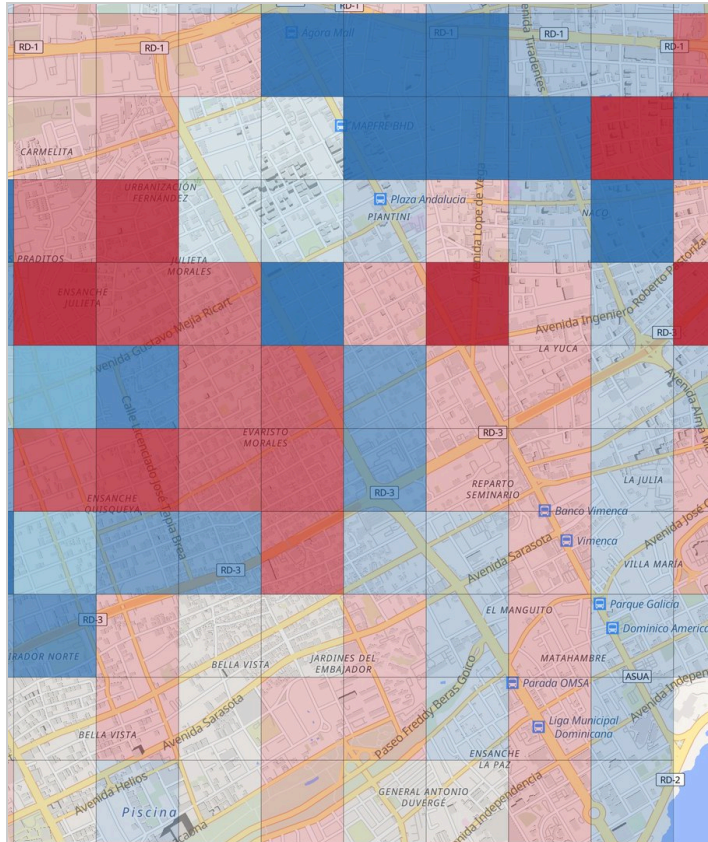
Learning curve



F1 across experiments

Santo Domingo		91.4%
No roof, code & year		93.2%
Quisquella -> Naco		94.8%
Leave-one-out no prior		79.0%
Leave-one-out with prior		82.8%

Conclusion



References

- Brzev, S., Scawthorn, C., Silva, V., et al. (2013). *GEM building taxonomy version 2.0*. <https://doi.org/10.13117/GEM.EXP-MOD.TR2013.02>
- GEM Foundation. (n.d.). *Dominican Republic exposure model* [Data set]. OpenQuake Global Risk Model. https://docs.openquake.org/global_risk_model/exposure/Caribbean_Central_America/Dominican_Republic/README.html
- GeomaticsCaminosUPM. (n.d.). *footprint_attributes* [Computer software]. GitHub. https://github.com/GeomaticsCaminosUPM/footprint_attributes
- Lopes, J., Gouveia, F., Silva, V., Moreira, R. S., Torres, J. M., Guerreiro, M., & Reis, L. P. (2023). Using deep learning for building stock classification in seismic risk analysis. In *Progress in Artificial Intelligence* (Lecture Notes in Computer Science, pp. 523–534). Springer. https://doi.org/10.1007/978-3-031-49011-8_41
- Marconcini, M., Esch, T., et al. (2021). Understanding current trends in global urbanisation: The World Settlement Footprint suite. *GI_Forum*. https://doi.org/10.1553/giscience2021_01_s33
- Torres, Y., Martínez-Cuevas, S., et al. (2023). Using remote sensing for exposure and seismic vulnerability evaluation: Is it reliable? *International Journal of Remote Sensing*. <https://doi.org/10.1080/15481603.2023.2196162>
- Ureña-Pliego, M., Rodríguez-Saiz, J., Núñez-Álvarez, G., Marchamalo-Sacristán, M., & González-Rodrigo, B. (2026). A methodology for the automated estimation of footprint-derived seismic behaviour modifiers in building exposure assessment. *Advanced Modeling and Simulation in Engineering Sciences*, 13(1), Article 3. <https://doi.org/10.1186/s40323-026-00323-y>
- Yepes-Estrada, C., Calderon, A., et al. (2023). Global building exposure model for earthquake risk assessment. *Earthquake Spectra*. <https://doi.org/10.1177/87552930231194048>
- Zhu, X. X., Chen, S., Zhang, F., Shi, Y., & Wang, Y. (2025). GlobalBuildingAtlas: An open global and complete dataset of building polygons, heights and LoD1 3D models. *Earth System Science Data*, 17(12), 6647–6668. <https://doi.org/10.5194/essd-17-6647-2025>



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Thank you

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